

Final Exam Practice Questions – Robotic Embedded Programming

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PART A

Q1. (MCQ)

In URDF, which element is used to specify the *visual appearance* of a link?

- A. <collision>
- B. <geometry>
- C. <visual>
- D. <sensor>

Q2. (MCQ)

Which ROS 2 tool is most commonly used to visualize robot models and sensor data?

- A. rqt_plot
- B. RViz2
- C. ros2 topic list
- D. tf2_echo

Q3. (Multi-Select)

Which of the following are advantages of using simulation before running a real robot?

- ☐ Identify configuration errors
- ☐ Automatically tune controllers
- ☐ Visualize sensor data
- ☐ Reduce risk of hardware damage

[Download Alternative Formats](#)**Q4. (Matching)**

Match the ROS 2 concept with its purpose:

Concept	Purpose
1. Topic	A. On-demand request/response
2. Service	B. Continuous data stream
3. Action	C. Long-running task with feedback

PART B**Q5.**

A robot in simulation appears to “sink” halfway into the floor.

List **two** possible causes that do *not* involve ros2_control.

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Q6.

Describe one real-world application where an IMU is essential for mobile robot operation and explain why.

Q7.

A ROS 2 node reads data from a temperature sensor and publishes it to `/temp_data`.

Give an example of what another node might do with this information.

Q8.

Explain why it is important to keep the URDF frame hierarchy consistent when adding new sensors.